

PETERHEAD HARBOUR, United Kingdom

WMO: 030920

Lat: 57.502N

Lon: 1.774W

Elev: 15

StdP: 101.14

Time Zone: 0.00 (EUW)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.3	0.7	-4.0	2.7	2.3	-3.1	2.9	2.5	15.0	5.8	14.2	5.2	4.4	270	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.6	18.1	14.9	17.1	14.2	16.3	13.9	15.7	17.1	15.1	16.3	14.6	15.7	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	16.4	14.5	10.4	15.8	14.0	10.0	15.2	44.1	17.3	42.4	16.3	40.9	15.8	18.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	12.0	10.8	DB	-2.1	20.9	1.7	1.6	-3.4	22.0	-4.4	23.0	-5.4	23.9	-6.6	25.1	(4)
(5)			WB	-2.9	17.5	1.6	0.8	-4.0	18.1	-4.9	18.6	-5.8	19.0	-6.9	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	5.4	5.2	6.0	7.5	9.5	11.9	13.8	14.2	12.8	10.4	8.0	5.9
	DBStd	3.76	1.88	2.30	2.25	1.94	1.90	1.62	1.43	1.45	1.69	2.19	2.09	2.56
	HDD10.0	730	141	135	126	78	31	2	0	0	2	23	64	128
	HDD18.3	3319	399	367	383	325	273	193	142	128	166	247	309	386
	CDD10.0	453	0	1	1	3	17	59	117	130	86	34	4	1
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	6.3	6.2	5.7	4.9	4.1	4.2	3.4	3.5	4.6	5.6	5.8	5.9
Precipitation	PrecAvg	781	64	60	55	51	54	58	62	57	66	96	90	73
	PrecMax	993	121	112	122	132	117	103	129	103	217	186	177	125
	PrecMin	565	27	13	18	23	13	13	20	14	16	16	56	25
	PrecStd	110	26	29	24	29	27	28	38	24	42	43	35	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	13.4	13.3	15.0	16.3	18.0	19.8	19.7	18.2	16.2	13.9	12.0
		MCWB	9.5	10.8	10.2	10.4	12.4	14.4	16.5	16.0	14.7	13.2	12.3	10.2
	2%	DB	9.9	11.3	11.3	12.4	14.3	16.2	17.8	18.2	16.9	14.5	12.5	10.7
		MCWB	8.3	9.4	8.6	9.3	11.0	12.9	15.0	15.0	14.1	12.3	10.6	9.3
	5%	DB	8.9	9.6	10.1	11.3	13.1	15.2	16.7	17.2	16.0	13.7	11.7	9.9
		MCWB	7.6	8.0	7.8	8.7	10.3	12.1	14.1	14.4	13.6	12.0	10.3	8.7
	10%	DB	8.2	8.4	9.0	10.3	12.1	14.2	15.8	16.4	15.1	13.1	10.9	9.3
		MCWB	7.1	6.7	7.1	8.1	9.8	11.7	13.8	14.2	13.3	11.6	9.6	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.2	10.6	11.5	12.8	15.0	17.0	16.8	15.7	13.8	12.4	10.5
		MCDB	11.2	12.9	12.6	13.8	15.6	17.1	19.1	18.7	17.1	15.1	13.1	11.4
	2%	WB	8.7	9.6	9.2	10.0	11.7	13.7	15.6	15.9	14.7	13.1	11.2	9.6
		MCDB	9.6	11.1	10.9	11.7	13.6	15.4	16.9	17.2	16.0	13.9	12.0	10.4
	5%	WB	7.9	8.2	8.1	9.2	11.0	12.9	14.9	15.3	14.2	12.5	10.6	8.9
		MCDB	8.6	9.4	9.5	10.6	12.4	14.3	16.0	16.4	15.2	13.3	11.4	9.7
	10%	WB	7.2	7.0	7.4	8.5	10.3	12.3	14.3	14.8	13.7	12.0	9.8	8.2
		MCDB	8.1	8.1	8.5	9.8	11.6	13.5	15.3	15.7	14.7	12.8	10.7	9.1
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.5	3.6	3.6	3.7	3.7	3.3	3.6	3.4	3.3	3.3	3.1
		MCDBR	4.1	5.0	5.5	5.6	5.3	5.2	4.8	4.9	4.6	4.0	3.9	3.5
		MCWBR	3.9	4.3	3.9	3.5	3.3	3.1	3.1	3.2	3.3	3.4	3.9	3.5
	5% WB	MCDBR	3.6	4.8	5.0	4.6	4.4	4.3	4.1	4.2	3.5	3.3	3.4	3.4
		MCWBR	3.6	4.5	3.8	3.3	2.9	3.1	2.9	3.0	3.0	3.2	3.6	3.6
Clear-Sky Solar Irradiance	taub	0.315	0.339	0.362	0.386	0.377	0.378	0.383	0.372	0.358	0.343	0.319	0.298	
	taud	2.240	2.324	2.329	2.292	2.378	2.405	2.398	2.437	2.474	2.416	2.282	2.177	
	Ebn at Noon	534	698	785	824	859	862	848	833	790	688	525	430	
	Edh at Noon	51	72	93	112	109	108	107	96	80	65	48	40	
All-Sky Solar Radiation	RadAvg	0.41	1.03	2.20	3.70	5.06	4.98	4.84	3.88	2.60	1.33	0.54	0.28	
	RadStd	0.05	0.10	0.21	0.34	0.27	0.39	0.34	0.26	0.21	0.11	0.05	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page